DISEASES

Chikungunya

Focusing on Odisha, the key symptoms of chikungunya reported in the region are:

- Sudden high fever lasting several days
- Severe joint pain affecting wrists, ankles, fingers, and toes, often causing difficulty in movement
- Joint swelling and stiffness, sometimes persisting for weeks or months
- Muscle pain and body aches
- Headache and fatigue common during the illness
- Rash appearing on the trunk and limbs
- Occasional nausea and vomiting
- Some cases report red eyes (conjunctivitis)

These symptoms usually begin 2 to 7 days after being bitten by the Aedes mosquito, which breeds extensively in Odisha's monsoon and post-monsoon seasons. The prolonged joint pain is a major cause of disability during outbreaks in Odisha, particularly affecting agricultural and working-age adults. Early diagnosis and symptomatic treatment are emphasized in Odisha’s public health response to manage outbreaks effectively.

Preventive Measures for Chikungunya

- Eliminate mosquito breeding sites by removing stagnant water in containers, pots, and drains.

- Use mosquito repellents and wear protective clothing (long sleeves and pants).
- Install mosquito nets or screens on windows and doors.
- Use mosquito coils or electric vaporisers indoors.
- Promote community awareness about mosquito control.
- Encourage early diagnosis and isolation of infected individuals to reduce mosquito infection

Treatment:

- No specific antiviral treatment available
- Rest and plenty of fluids to prevent dehydration
- Over-the-counter pain relievers like acetaminophen to reduce fever and joint pain (avoid NSAIDs initially)
- Medical care for severe joint pain or complications

Home Remedies:

- Drink coconut water to stay hydrated and support liver function
- Turmeric and ginger for their anti-inflammatory and pain-relief properties
- Epsom salt baths to reduce joint pain
- Neem leaf decoction baths to soothe rashes and inflammation
- Papaya leaf juice to potentially boost platelet count
- Apply cold compresses on painful joints
- Use natural mosquito repellents and eliminate stagnant water to prevent infection

CHOLERA

The most prone symptoms of cholera in Odisha are:

- Sudden onset of profuse watery diarrhea, often described as "rice water" stools (milky, pale, with a fishy smell)
- Frequent diarrhea, sometimes producing up to 1 liter of stool per hour
- Vomiting, often intense, occurring early in the illness
- Rapid dehydration signs: extreme thirst, dry mouth, sunken eyes, decreased urine output, and dry skin
- Fatigue and irritability, especially in children
- Muscle cramps due to electrolyte imbalance (loss of salts like sodium, potassium)
- Restlessness or lethargy
- Low blood pressure and weak, rapid pulse in severe cases
- Skin that loses elasticity (slow to return after pinching)
- In severe untreated cases, shock, kidney failure, and death may occur rapidly

Preventive Measures for Cholera

- Drink only safe, treated, or boiled water.
- Ensure proper sanitation and use of toilets to avoid open defecation.

- Wash hands thoroughly with soap and clean water before eating and after using the toilet.
- Avoid raw or undercooked food, especially seafood.
- Use safe methods for food preparation and storage.
- Prompt treatment of cholera cases to prevent spread via contaminated stools.
- Implement community health education on hygiene practices.

Treatment:

- Immediate rehydration with Oral Rehydration Salts (ORS) or intravenous fluids in severe cases
- Antibiotics prescribed by a doctor in some cases
- Continuous monitoring of electrolytes and hydration levels

Home Remedies:

- Homemade ORS (mix 1/2 tsp salt + 6 tsp sugar in 1 liter clean water)
- Drink plenty of fluids like coconut water, lime water
- Herbal remedies like ginger juice mixed with black pepper, holy basil, and mint
- Probiotic yogurt to restore gut flora
- Clove and black pepper boiled in water as a soothing drink
- Proper sanitation and hygiene to prevent spread

DIARRHEAL DISEASES

The most common symptoms of diarrheal diseases in Odisha are:

- Frequent loose or watery stools
- Abdominal cramps or pain
- Nausea and vomiting
- Fever
- Dehydration signs like dry mouth, excessive thirst, and reduced urination
- Fatigue and general weakness
- Urgency to pass stools
- Sometimes bloating and gas
- In severe cases, irritability, confusion, dizziness, and skin dryness

Preventive Measures for Diarrheal Diseases

- Practice good hand hygiene with soap and water, especially before meals and after using the toilet.
- Drink safe, hygienic water—boil, filter, or treat water before drinking.
- Maintain clean and hygienic food handling and cooking practices.
- Improve sanitation and proper disposal of human waste.
- Promote breastfeeding for infants to improve immunity.
- Keep infants and children away from contaminated water sources.

- Seek timely medical care for diarrheal symptoms to manage dehydration.

Treatment:

- Maintain hydration with fluids and ORS
- Seek medical care if severe or prolonged symptoms occur
- Use antibiotics only if bacterial infection is confirmed

Home Remedies:

- BRAT diet (Bananas, Rice, Applesauce, Toast) to soothe the gut
- Drink rice water to restore nutrients and calm intestines
- Herbal tea such as chamomile, fennel, and ginger tea for antispasmodic and anti-inflammatory effects
- Lemon water with a pinch of salt and sugar for electrolyte balance
- Turmeric mixed in warm milk for antibacterial and healing effects
- Psyllium husk to firm up stools and reduce irritation
- Fenugreek seeds powder mixed in water to relieve abdominal discomfort
- Avoid fatty, spicy, or heavy foods during illness

MALARIA

* *Symptoms:*

* *Classic "Malaria Attack":* A cyclical occurrence of high fever, chills, and profuse sweating.

* *Other Symptoms:* Headache, muscle and joint pain, fatigue, nausea, vomiting, and diarrhea.

* *Severe Symptoms:* In severe cases, particularly with the P. falciparum parasite, it can lead to cerebral malaria (seizures, coma), severe anemia, and organ failure.

* *Prevention:*

* *Vector Control:* The primary method is to avoid mosquito bites by using insecticide-treated bed nets (ITNs), insect repellent, and protective clothing.

* *Prophylactic Drugs:* Travelers to endemic areas may take antimalarial medication to prevent infection.

* *Vaccines:* The *RTS,S/AS01* and *R21/Matrix-M* vaccines are now recommended for children in high-risk areas.

* *Treatment:*

* Malaria is a medical emergency and requires prompt treatment with antimalarial drugs. The type of medication depends on the specific parasite, the patient's age, and the severity of the illness.

* *Artemisinin-based combination therapies (ACTs)* are the standard treatment for uncomplicated P. falciparum malaria.

* Drug resistance is a growing challenge, and research is ongoing to develop new therapies.

DENGUE

* *Symptoms:*

  * *Sudden High Fever:* Often accompanied by a severe headache and pain behind the eyes.

  * *"Breakbone Fever":* Severe muscle and joint pain that can be debilitating.

  * *Other Symptoms:* Nausea, vomiting, and a skin rash.

  * *Severe Dengue (Dengue Hemorrhagic Fever/Dengue Shock Syndrome):* A medical emergency with warning signs like severe abdominal pain, persistent vomiting, and bleeding.

* *Prevention:*

  * *Vector Control:* The main focus is to control the Aedes aegypti mosquito population. This includes eliminating standing water where mosquitoes breed and using repellents.

  * *Vaccines:* The *Qdenga (TAK-003)* vaccine is approved in some regions to protect against the four dengue serotypes.

* *Treatment:*

  * There is no specific antiviral cure for dengue. Treatment is supportive and focuses on managing symptoms.

  * *Fluid Management:* Drinking plenty of fluids is crucial to prevent dehydration. In severe cases, intravenous (IV) fluids may be required in a hospital setting.

* *Pain Relief:* Use acetaminophen (paracetamol) for fever and pain. *Avoid NSAIDs* like ibuprofen and aspirin as they can increase the risk of bleeding.

TUBERCULOSIS

Symptoms of Tuberculosis in Odisha

- Persistent cough lasting more than 2 weeks
- Chest pain
- Coughing up blood (hemoptysis)
- Unexplained weight loss
- Fever, often low-grade and sweats at night
- Fatigue and general weakness
- Loss of appetite
- Swelling of lymph nodes in some cases

Preventive Measures

- Early diagnosis and treatment adherence to stop spread
- Ensure proper ventilation in living and working environments
- Avoid close contact with active TB patients until they are treated
- BCG vaccination for infants as part of immunization program
- Maintain good nutritional status to strengthen immunity
- Public awareness and education on cough hygiene
- Prompt screening of high-risk groups, including tribal populations

- Improve living conditions and reduce overcrowding

Home Remedies for Tuberculosis

- Nutritious Diet:
    - Include protein-rich foods like eggs, lentils, lean meat, and dairy products to help repair damaged tissues.
    - Vitamin C-rich foods such as oranges, lemons, amla (Indian gooseberry), and bell peppers to boost immunity.
    - Foods rich in zinc and selenium (nuts, seeds, whole grains) support healing and immune response.
- Herbal Remedies:
    - Garlic: Contains antibacterial properties and helps activate the lymphatic system to reduce inflammation. Consume 1-2 cloves daily after consulting a doctor.
    - Turmeric Milk: Anti-inflammatory effect aids lung recovery—add a pinch of turmeric to warm milk and drink regularly.
    - Tulsi (Holy Basil) Tea: Supports respiratory health and boosts immunity.
    - Black Pepper and Honey Mixture: Soothes cough and throat irritation.
- Hydration and Air:
    - Drink plenty of water, herbal teas, and warm soups to loosen mucus and flush out toxins.
    - Spend 15-20 mins daily in mild morning sunlight to get vitamin D which strengthens immunity.

    - Ensure proper ventilation in living spaces to reduce infection risk.
- Rest and Stress Reduction:
    - Adequate sleep and rest are crucial for recovery.
    - Gentle physical exercises like walking or yoga improve lung function.
    - Avoid smoking and alcohol as they weaken lung and immune health.
- Specific Traditional Remedies:
    - Amla (Indian gooseberry) juice mixed with honey: Drink regularly to support recovery.
    - Mixture of drumstick leaves, black pepper powder, salt, and lemon juice in hot water: Used as a detoxifying concoction.
    - Walnut, garlic, and ghee paste: Consume once daily for lung strength.
    - Black pepper and asafoetida fried in ghee: Taken in doses thrice a day to reduce lung inflammation

Treatment

- WHO-recommended multi-drug anti-tuberculosis therapy (typically 6 months)
- Directly Observed Treatment, Short-course (DOTS) for ensuring adherence
- Regular monitoring by healthcare providers for side effects and progress

- Treatment is free and available through government health programs
- Management of drug-resistant TB with specialized care if needed
- Supportive care for nutrition and managing symptoms during treatment

HEPATITIS

* *Symptoms:*

  * *Acute Hepatitis:* Symptoms can be flu-like and include fatigue, fever, nausea, and abdominal pain. Jaundice (yellowing of the skin and eyes) is a key sign.

  * *Chronic Hepatitis:* Often asymptomatic for many years. It can silently progress to severe liver damage, including *cirrhosis* (liver scarring) and *liver cancer*.

* *Prevention:*

  * *Hepatitis A & B:* Vaccines are highly effective and are the best way to prevent these infections.

  * *Hepatitis A & E:* Can be prevented by good hygiene, sanitation, and consuming safe food and water.

  * *Hepatitis B & C:* Prevention involves avoiding contact with infected blood and bodily fluids. This includes not sharing needles, practicing safe sex, and ensuring sterilized equipment for medical procedures, tattoos, and piercings.

* *Treatment:*

* *Acute Hepatitis:* Treatment is supportive, focusing on rest, nutrition, and hydration.

* *Chronic Hepatitis B:* While not curable, antiviral medications can effectively manage the virus, preventing further liver damage.

* *Chronic Hepatitis C:* This is a major success story. It is now curable in over 95% of cases using short-course, oral *direct-acting antiviral (DAA)* medications. The main challenge now is widespread screening to find those who are infected and providing access to treatment.

Home Remedies and Supportive Care

- Rest: Adequate rest is essential to help the liver recover during infection.
- Hydration: Drink plenty of fluids like water, coconut water, and fresh fruit juices to stay hydrated and support liver function.
- Nutritious Diet: Consume easily digestible, nutrient-rich foods such as fruits, vegetables, whole grains, and lean proteins. Avoid oily, spicy, and processed foods.
- Ayurvedic Herbs:
    - *Giloy (Tinospora cordifolia):* Known for immune boosting and liver-protective properties, commonly used in Odisha and Ayurveda for viral hepatitis.
    - *Kutki (Picrorhiza kurroa):* Helps detoxify and regenerate liver cells.
    - *Triphala:* Acts as an antioxidant and promotes digestion.

    - *Indian Aloe Vera:* Rich in antioxidants and nutrients supporting liver health.
    - *Bhumi Amalaki:* Detoxifies and nourishes liver.
- **Avoid Alcohol and Smoking:** Both worsen liver damage and slow recovery.
- **Moderate Exercise:** After acute symptoms improve, light physical activity helps enhance overall health.
- **Avoid Certain Herbal Supplements Without Supervision:** Some herbs can interact with medications or harm the liver.

JAPANESE ENCEPHALITIS (JE)

Japanese encephalitis is a viral infection of the brain, transmitted by mosquitoes, primarily in rural parts of Asia. It is a significant public health concern, especially among children.

* *Symptoms:*

* Most infections are *asymptomatic*, meaning they show no symptoms at all.

* In a small percentage of cases, a severe infection develops, leading to *encephalitis* (brain swelling).

* *Initial symptoms* are flu-like: high fever, headache, and vomiting.

* *Neurological symptoms* follow: neck stiffness, disorientation, tremors, convulsions, muscle weakness, and, in severe cases, paralysis and coma.

* A high percentage of survivors are left with permanent neurological or psychiatric disabilities.

* *Advanced Steps & Treatment:*

* *No Specific Cure:* There is no antiviral drug available to cure JE once it has developed.

* *Supportive Care:* The sole approach is supportive treatment to manage symptoms and complications. This includes:

    * Medications to control fever and seizures.

    * Respiratory support and fluid management for patients in critical condition.

    * Managing pressure inside the skull (intracranial pressure) from brain swelling.

* *Early Diagnosis:* Advancements in diagnostics, like rapid tests and molecular assays, allow for faster identification of the virus, which is crucial for initiating supportive care and managing outbreaks.

* *Prevention & Control:*

* *Vaccination:* This is the most effective and a cornerstone of JE prevention. Several safe and effective vaccines are available globally, including inactivated and live-attenuated vaccines.

* *Vector Control:* Since mosquitoes transmit the virus, personal protection is vital. This involves:

    * Using insect repellent.

    * Wearing long-sleeved clothing.

* Using mosquito nets.

* *Innovative Research:* Ongoing research is exploring new vaccine candidates, including plant-based vaccines, and improved vector control strategies.

Supportive Home Remedies

- Ensure complete bed rest to reduce strain and support recovery.
- Maintain good hydration—give fluids like water, coconut water, clear soup, and ORS if the patient is conscious and able to drink.
- Mild pain and fever may be managed with paracetamol (acetaminophen) as advised by a doctor; avoid NSAIDs due to bleeding risks.
- Use cool compresses or light clothing to help reduce fever.
- Provide easily digestible, nutritious foods if the patient can eat—such as fruits, boiled rice, and vegetables.
- Use mosquito nets and repellents to prevent further bites to the patient or household members.
- Keep the patient's environment calm and minimize noise to avoid unnecessary stress on the brain.

LEPTOSPIROSIS

Leptospirosis is a bacterial disease spread through contact with the urine of infected animals, primarily through contaminated water or soil. It is a widespread zoonotic disease.

* *Symptoms:*

  * *Biphasic Illness:* The disease often occurs in two phases.

    * *First Phase:* Sudden onset of flu-like symptoms, including high fever, severe headache, muscle aches (especially in the calves), and chills.

    * *Second Phase:* After a brief period of feeling better, some people can develop a more severe form.

  * *Severe Leptospirosis (Weil's Disease):* A life-threatening condition characterized by:

    * Jaundice (yellowing of the skin and eyes) from liver damage.

    * Kidney failure.

    * Meningitis (inflammation of the brain and spinal cord lining).

    * Hemorrhage (bleeding), particularly in the lungs.

* *Advanced Steps & Treatment:*

  * *Antibiotics:* Unlike JE, Leptospirosis is treatable with antibiotics, especially when administered early.

  * *Mild Cases:* Can be treated with oral antibiotics like *doxycycline* or *amoxicillin*.

  * *Severe Cases:* Require hospitalization and intravenous (IV) antibiotics such as *penicillin G* or *ceftriaxone*.

  * *Supportive Care:* In severe cases, supportive care is crucial to manage complications like kidney failure (dialysis may be needed) and lung bleeding.

* *Prevention & Control:*

* *Avoid Contaminated Water:* The best way to prevent infection is to avoid swimming or wading in freshwater that might be contaminated with animal urine, especially after heavy rainfall or floods.

* *Protective Equipment:* Individuals in high-risk occupations (e.g., farmers, sewer workers) should wear protective clothing and footwear.

* *Vaccines:* Human vaccines are available in some countries but are not widely used globally, as they often provide protection against only a limited number of the many serovars (types) of the bacteria. Research is ongoing to develop a broader vaccine.

* *Animal Vaccination:* Vaccinating domestic and livestock animals, particularly dogs and cattle, is an effective strategy to reduce the spread of the disease.

Supportive Home Remedies

- **Rest:** Take plenty of rest to allow the body to recover and avoid strenuous activity until symptoms resolve.
- **Hydration:** Drink plenty of fluids (water, clear soup, coconut water) to prevent dehydration, especially if fever or vomiting is present.
- **Nutrition:** Eat nutritious, easily digestible foods rich in proteins, vitamins, and minerals. Foods like dal, rice, boiled vegetables, and fruits are helpful.

- **Turmeric:** Turmeric (haldi) has anti-inflammatory and antimicrobial properties. Mix a pinch into warm milk and drink twice daily.
- **Ginger:** Ginger helps reduce inflammation and may support healing. Drink ginger tea made by boiling fresh ginger in water.
- **Milk Thistle:** This herbal supplement may support liver health, but consult a doctor before use.
- **Pain and Fever Relief:** Take paracetamol or acetaminophen (consult a doctor before using other painkillers).
- **Avoid Alcohol and Smoking:** Both can worsen organ damage during infection.
- **Monitor Symptoms:** If symptoms worsen (jaundice, severe weakness, decreased urination, confusion), seek immediate medical help.

COMMON DISEASES ALL OVER INDIA

1. *Common Cold and Influenza (Flu)

These are frequent viral infections of the respiratory system that spread very easily, especially during the monsoon and winter.

Symptoms

- Runny or stuffy nose
- Sneezing
- Sore or scratchy throat
- Cough

- Mild headache and body aches
- Fatigue and general malaise
- Low-grade fever (more common in children)
- Watery eyes
- In influenza, symptoms can be more severe with sudden onset of high fever, chills, muscle pain, and fatigue lasting several days, sometimes progressing to pneumonia.

Home Remedies

- **Ginger Tea (Adrak Chai):** Boil fresh ginger in water, strain, add honey, drink 2-3 times daily to soothe throat and reduce congestion.
- **Turmeric Milk (Haldi Doodh):** Warm milk with turmeric helps reduce inflammation and boost immunity.
- **Ajwain Steam Inhalation:** Boil water with carom seeds and inhale steam to clear nasal passages.
- **Honey with Black Pepper:** A spoon of honey mixed with black pepper relieves cough and soothes throat.
- **Spiced Kadha:** A herbal drink made with cloves, cinnamon, tulsi, black pepper, and ginger that acts as an immunity booster and toxin eliminator.
- **Rest and Hydration:** Plenty of rest and fluids including warm water and herbal teas promote recovery.

Preventive Measures

- Frequent hand washing with soap and water.
- Avoid close contact with people showing cold or flu symptoms.

- Cover mouth and nose while sneezing or coughing using tissue or elbow crease.
- Use masks or cover face in crowded or enclosed spaces, especially during flu outbreaks.
- Keep living and working areas well ventilated.
- Strengthen immunity through balanced diet, adequate sleep, and regular exercise.
- Seasonal flu vaccination is available and recommended for vulnerable groups.

2. *DENGUE AND MALARIA*

These are vector-borne diseases transmitted by mosquitoes that breed in stagnant water, making them most common during and after the monsoon season.

* *Symptoms:*

* *Dengue:* High fever, severe headache, intense joint and muscle pain ("breakbone fever"), and a skin rash. Severe cases can lead to dangerous bleeding.

* *Malaria:* Recurrent episodes of high fever with shaking chills and sweating. Other symptoms include headache, body aches, and fatigue.

* *Prevention:*

* *Vector Control:* Eliminate stagnant water around your home. Cover water containers and discard old tires or pots.

* *Personal Protection:* Use mosquito nets, wear long-sleeved clothes, and apply insect repellent on exposed skin.

* *Home Remedies:*

Dengue Home Remedies

- Papaya Leaf Juice: Widely used in Odisha, believed to help increase platelet count and support recovery. Crush fresh papaya leaves, extract the juice, and consume twice daily.
- Giloy Juice: Known for its immune-boosting and anti-inflammatory properties. Boil giloy stems and drink the water to improve immunity.
- Neem Leaves: Used for their antibacterial and antiviral properties; neem twigs can be chewed or neem leaf juice consumed.
- Tulsi (Basil) Tea: Helps reduce fever and boosts immune defenses. Boil tulsi leaves and drink the tea with a little honey.
- Barley Water: Hydrating and nutritious, supports digestion and recovery.
- Hydration: Drink plenty of fluids like water, coconut water, and fresh juices to prevent dehydration during fever.
- Rest: Complete rest is essential for recovery.
- Avoid: Aspirin and NSAIDs, which can worsen bleeding risks.

Malaria Home Remedies

- Ginger Water: Boil ginger in water and drink 1-2 cups daily to reduce symptoms.
- Turmeric Milk: Antioxidant and anti-inflammatory, helps detoxify and reduce pain.

- **Cinnamon and Honey:** Boil cinnamon and black pepper in water; add honey and drink to fight malaria parasites.
- **Fresh Juices:** Orange juice, sweet lime juice, or grapefruit juice to boost immunity and provide vitamins.
- **Herbal Teas:** Herbal teas with spices and herbs to reduce fever and fatigue.
- **Apple Cider Vinegar:** Mixed with water, soaked in a towel, and applied on calves to reduce fever.
- **Use mosquito nets and repellents:** Protect against mosquito bites, especially during monsoon.
- **Keep surroundings clean:** Remove stagnant water to prevent mosquito breeding.

3. *TYPHOID AND CHOLERA

These are bacterial diseases spread through contaminated food and water, which is a common problem during the monsoon season.

* *Symptoms:*

* *Typhoid:* A prolonged high fever that rises in a "stepwise" manner, abdominal pain, headache, and weakness.

* *Cholera:* Sudden, severe watery diarrhea and vomiting, which can lead to life-threatening dehydration very quickly.

* *Prevention:*

* *Clean Water & Food:* Only drink boiled, filtered, or bottled water. Avoid uncooked food, especially from unhygienic street vendors.

* *Hygiene:* Wash hands thoroughly with soap, especially before eating and after using the toilet.

* *Vaccination:* Typhoid and cholera vaccines are recommended for those living in or traveling to high-risk areas.

* *Home Remedies:*

* *Oral Rehydration Solution (ORS):* For cholera and other diarrheal diseases, a simple ORS made from sugar and salt in clean water is life-saving for preventing dehydration.

Supportive Home Remedies

- **Hydration:** Drink plenty of fluids—water, coconut water, fruit juices, homemade soups, and Oral Rehydration Solution (ORS). Hydration combats fever-related dehydration and helps flush toxins.
- **Light, Bland Diet:** Eat soft, easy-to-digest foods such as rice, mashed potatoes, porridge (oats or suji/dalia), khichdi, boiled vegetables (carrots, bottle gourd, pumpkin), bananas, and stewed apples. These foods provide energy and are gentle on the gut.
- **Probiotics:** Consume yogurt, curd, and buttermilk to support gut health and immunity.
- **Frequent, Small Meals:** Take small portions at regular intervals to avoid straining digestion and help regain strength.
- **Herbs:** Basil leaves (tulsi)—boil in water, strain, and drink to calm the stomach. Ginger tea, mint tea, and herbal teas may soothe digestion.
- **Banana:** Contains pectin and potassium, helps control diarrhea and replace electrolytes.

- **Cloves:** Boil water with cloves, strain and sip occasionally to fight bacteria and ease nausea.
- **Pomegranate:** Juice or seeds aid hydration and provide antioxidants.
- **Rest:** Get plenty of rest and avoid strenuous activity; helps the body recover from weakness and fatigue.
- **Sponging:** If fever is high, sponge the body with lukewarm water to help bring down temperature.

4. *NON-COMMUNICABLE (LIFESTYLE) DISEASES*

Diseases like *diabetes, hypertension, and cardiovascular diseases* are increasingly common due to changing lifestyles, unhealthy diets, and reduced physical activity.

* *Symptoms:*

  * *Diabetes:* Frequent urination, excessive thirst and hunger, fatigue, and unexplained weight loss.

  * *Hypertension (High Blood Pressure):* Often called a *"silent killer"* as it has no symptoms.

  * *Cardiovascular Diseases:* Chest pain, shortness of breath, and pain in the arm, neck, or jaw.

* *Prevention:*

  * *Healthy Diet:* Eat a balanced diet with lots of fresh fruits, vegetables, and whole grains. Reduce your intake of processed foods, sugar, and unhealthy fats.

* *Physical Activity:* Regular exercise, like walking, jogging, or yoga, for at least 30 minutes a day is crucial.

* *Lifestyle Changes:* Avoid smoking and limit alcohol. Manage stress through activities like meditation.

Diabetes Home Remedies

- **Ajwain Water:** Boil ajwain (carom seeds), fennel seeds, and cinnamon powder in water and drink after meals to help control blood sugar levels.
- **Aloe Vera Juice:** May help regulate insulin and improve blood sugar management.
- **Amla (Indian Gooseberry):** Rich in vitamin C and chromium, aids in improving insulin sensitivity and blood sugar control. Consume as fresh fruit, juice, or powder.
- **Jamun Seeds:** Contains compounds that slow glucose release; grind and consume with water.
- **Fenugreek Seeds:** Soaked or sprouted seeds included in diet help reduce blood glucose.
- **Bitter Gourd (Karela):** Consumed as juice or cooked vegetable, helps lower blood sugar.
- **Neem Leaves:** Drinking neem extract or consuming leaves supports blood sugar balance.
- **Stress Management and Exercise:** Regular physical activity and stress reduction techniques improve sugar control.

Hypertension (High Blood Pressure) Home Remedies

- **Coriander and Cardamom Drink:** Take coriander and cardamom infused fresh peach juice 2-3 times daily to mildly reduce blood pressure.
- **Moong Dal Soup:** Rich in nutrients, lowers high blood pressure when consumed regularly.
- **Honey Water with Apple Cider Vinegar:** In the morning, helps regulate blood pressure and improve circulation.
- **Watermelon with Cardamom and Coriander:** Acts as a mild diuretic helping reduce blood pressure.
- **Reduce Salt Intake:** Avoiding salty, oily, and fried foods is crucial.
- **Herbal Tea:** Mixture of punarnava, passion flower, and hawthorn berry steeped in hot water supports blood pressure control.
- **Physical Activity and Weight Management:** Regular exercise and maintaining healthy weight are vital.

Cardiovascular Diseases Home Remedies

- **Garlic:** Contains antioxidants and helps reduce cholesterol and blood pressure.
- **Ginseng and Hawthorn:** Traditional herbs used for improving heart health.
- **Green Tea (Camellia Sinensis):** Antioxidant effects beneficial for blood vessels.
- **Ginkgo Biloba and Rosemary:** Improve blood circulation and reduce oxidative stress.
- **Diet:** Balanced diet rich in fruits, vegetables, nuts, and omega-3 fatty acids supports heart health.

- **Stress Management:** Yoga, meditation, and breathing exercises improve cardiovascular function.

Mental Health Awareness in Odisha

- Mental health issues like depression, anxiety, and bipolar disorder are increasingly prevalent in Odisha, including rural and tribal populations. Over 36% of tribal respondents in some studies reported moderate depression, with poor socio-economic conditions and lack of education contributing factors.
- Odisha faces challenges such as stigma, superstitions, and low awareness, especially in tribal regions, hindering people from seeking professional help.
- The government and organizations are running awareness programs in colleges, universities, and communities to educate about mental health, stress management, suicidal tendencies, and coping skills.
- Tele-MANAS helpline (14416) and other mental health helplines provide counseling, but awareness of these services is still low among youth.

Common Mental Health Issues in Odisha

- Depression, anxiety, stress, substance abuse, and suicidal tendencies, especially in youth and tribal populations.
- Academic pressure and social/psychological stress are major triggers.
- Mental health is affected by socio-economic, cultural, and environmental factors.

Remedies and Support for Mental Health

- **Psychological therapies:** Cognitive Behavioural Therapy (CBT), Interpersonal Therapy (IPT), and other talk therapies help manage symptoms and develop coping skills.
- **Medication:** Antidepressants, anti-anxiety, mood stabilizers, and antipsychotic drugs as prescribed by psychiatrists for specific conditions.
- **Community Awareness and Education:** Reducing stigma by educating communities through street plays, workshops, and training frontline workers.
- **Counseling Services and Helplines:** Encouraging use of 24x7 helplines and tele-counseling for immediate help.
- **Social Support:** Family and community support play a crucial role in recovery and maintenance of mental well-being.
- **Healthy Lifestyle:** Regular exercise, balanced diet, adequate sleep, and mindfulness practices like yoga and meditation promote mental well-being.
- **Special Focus on Tribal Areas:** Addressing cultural beliefs and superstitions, improving education, and facilitating access to professional mental healthcare.

Government and NGO Initiatives

- MoU between Odisha government and Indian Psychiatric Society to conduct mental health awareness in educational institutions.
- Mental health rehabilitation programs and community therapy initiatives in tribal regions.
- Training of health workers to identify and manage common mental health disorders.

Mental health care in Odisha requires multi-sectoral efforts involving healthcare providers, educators, families, and communities to effectively tackle the increasing burden and help people live healthier, fulfilling lives.

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CAMPAIGNS

Here are detailed healthcare campaigns in Odisha including their time periods and benefits:

Sustha Nari, Sashakta Parivaar Campaign

- **Time Period:** Launched in September 2025
- **Benefits:**
    - Early detection and screening of over 29,000 women for cancers, menstrual disorders, and malnutrition.
    - Over 3,000 health camps conducted improving access to preventive care.
    - Use of digital tracking improves follow-up and management.
    - Empowers women through health education and community engagement.

    - Increases awareness on nutrition, hygiene, and women's health issues.

Vision 2025 Universal Healthcare

- **Time Period:** Ongoing vision with target by 2025
- **Benefits:**
    - Establishment of secondary care hospitals in each district enhances local access.
    - Improved health infrastructure and availability of advanced medical equipment.
    - Focus on equitable healthcare access, especially maternal and child health.
    - Encourages private sector participation to improve services.
    - Expands health insurance coverage reducing financial burden on patients.

Ayushman Bharat- Unified Health Insurance Scheme

- **Time Period:** Launched and operational as of 2024-2025
- **Benefits:**
    - Provides cashless treatment for over 35 million Odisha residents.
    - Coverage includes tertiary, secondary, and select primary care treatments.
    - Protection for vulnerable groups, including senior citizens.
    - Integrates multiple health programs, simplifying access.

    - Promotes empanelment of diverse health facilities increasing patient options.

Public-Private Partnerships (PPP)

- Time Period: Active under National Health Mission, Odisha (ongoing since 2020s)
- Benefits:
    - Expands healthcare outreach to rural and underserved urban areas.
    - Improves vaccination rates and disease surveillance.
    - Strengthens peripheral health services and workforce capacity.
    - Enhances health data reporting and monitoring systems.

Indian Institute of Public Health Bhubaneswar (IIPHB)

- Time Period: Established and actively contributing since early 2020s
- Benefits:
    - Builds public health workforce capacity through specialized training.
    - Provides research and innovative solutions tailored to Odisha's health needs.
    - Supports disease elimination programs and health system improvements.

9th National Health Summit at Puri, Odisha

- Time Period: Held in September 2025
- Benefits:

    - Platform for sharing effective public health practices nationally.
    - Accelerates disease elimination efforts like TB in Odisha.
    - Fosters multi-sector collaboration and innovation in healthcare delivery.

Biju Swasthya Kalyan Yojana

- **Time Period:** Operational since 2024-2025
- **Benefits:**
    - Universal health coverage providing free secondary and tertiary care in government hospitals.
    - Significantly reduces out-of-pocket expenses for marginalized populations.
    - Covers free medicines, diagnostics, surgeries, and emergency care.

National Health Mission (NHM) Odisha

- Umbrella program addressing maternal and child health, communicable and non-communicable diseases.
- Implements immunization drives, sanitation programs, and health awareness campaigns.
- Strengthens primary healthcare infrastructure and workforce training.
- Partners with NGOs and private sector under Public-Private Partnerships (PPP).

Odisha is a state highly vulnerable to natural disasters like cyclones, floods, droughts, and heatwaves. In recent years, it has also developed robust systems to manage epidemics. The state's disaster management strategy, spearheaded by the Odisha State Disaster Management Authority (OSDMA), emphasizes community participation and preparedness. Here are some general protocols and guidelines local people should follow during a natural calamity or an epidemic outbreak:

Protocols for Natural Calamities (e.g., Cyclones, Floods)

Odisha's disaster management approach has shifted from a "relief-centric" to a "proactive" one, with a focus on early warnings and community-based preparedness.

Before the Calamity:

- **Stay Informed:** Pay close attention to early warning signals and weather forecasts from the India Meteorological Department (IMD), OSDMA, and local government authorities. The state has an effective early warning system that disseminates information through sirens, mass messaging, and other channels.
- **Prepare a Disaster Kit:** Keep a kit with essential items, including:
    - Dry food, clean drinking water, and a water purification method.
    - First-aid supplies, essential medicines, and a list of medical conditions and allergies for family members.
    - A battery-operated radio, a flashlight with extra batteries, and a power bank for mobile phones.

    - Copies of important documents (ID cards, land deeds, insurance papers) in a waterproof bag.
    - Cash in small denominations.
    - Matches or a lighter in a waterproof container.
    - Basic tools and a whistle to signal for help.
- Secure Your Home:
    - Store important items and documents in a safe, high place to protect them from floodwaters.
    - Trim trees and branches that could fall on your house.
    - Secure or bring in outdoor items that could be blown away by strong winds.
    - If you live in a kutcha (mud and thatch) house, consider reinforcing it or moving to a safer location.
- Know Your Evacuation Plan:
    - Identify the nearest multipurpose cyclone/flood shelters.
    - Know the designated evacuation routes from your village.
    - Listen to instructions from local authorities (e.g., Sarpanch, police, volunteers) regarding evacuation.

During the Calamity:

- Evacuate when instructed: If authorities issue an evacuation order, do not hesitate to leave immediately and go to a designated shelter.
- Stay Indoors: If you are in a safe, multi-story building, stay indoors and away from windows and doors.

- **Disconnect Power and Gas:** Turn off the main electricity switch and close the gas cylinder valve to prevent fire and electric shocks.
- **Do Not Drive or Walk Through Water:** Floodwaters can be fast-moving and contain dangerous debris or hidden hazards. Do not attempt to cross flooded roads or bridges.
- **Stay Calm:** Panic can lead to poor decisions. Help others, especially the elderly, children, and people with disabilities, and follow the instructions of the authorities.

Protocols for Epidemic Outbreaks

Odisha's response to past epidemics, such as the COVID-19 pandemic, has been characterized by a decentralized, community-driven approach, empowering local leaders and volunteers.

General Guidelines:

- **Follow Health Advisories:** Stay updated on health advisories from the Health and Family Welfare Department, district health officials, and local government bodies.
- **Practice Good Hygiene:**
    - Wash hands frequently with soap and water or use an alcohol-based sanitizer.
    - Maintain social distancing as advised by health authorities.
    - Wear a face mask in public places, especially if instructed to do so.
- **Seek Medical Attention:**

    - If you or a family member develop symptoms, contact the designated health helpline (e.g., 104 or 14410 during COVID-19) for advice.
    - Do not self-medicate. Follow the instructions of doctors and health workers.
    - Go to a designated medical facility for testing and treatment if necessary.
- Cooperate with Local Authorities:
    - Support and cooperate with local committees and officials (e.g., Sarpanches, ASHA workers) in implementing containment measures.
    - Register if you are a returning traveler or have been in a high-risk area, and follow quarantine protocols as required.
- Rely on Authentic Information:
    - Avoid spreading or believing rumors and misinformation, especially through social media.
    - Get your information from reliable sources like government websites, press releases, and reputable news outlets.

Community-Level Actions:

- Volunteer and Assist: Local communities, including women's self-help groups and youth clubs, often play a crucial role in awareness campaigns, distributing essential supplies, and helping vulnerable groups.
- Enforce Regulations: Community members can help in enforcing guidelines like social distancing, mask-wearing, and hygiene

protocols at the local level, ensuring the safety of everyone in the village or neighborhood.

By adhering to these protocols, local people in Odisha can significantly reduce their risk and contribute to the collective resilience of the community during a crisis.

Emergency Transport & Ambulance Services

- **108 Emergency Medical Ambulance Service (EMAS):**
  - Free emergency ambulance (ALS, BLS, boat ambulances).
  - Covers trauma, accidents, maternal emergencies, children, etc.
  - Dial **108** (toll-free).
- **Janani Express:**
  - Free transport for pregnant women to hospitals.
  - Part of Integrated Patient Transport Service.
- **Boat Ambulances:**
  - For riverine / flood-prone / inaccessible areas.
- **104 Health Helpline:**
  - Medical advice, counseling, and referral guidance.

Health Coverage & Financial Assistance

- **Biju Swasthya Kalyan Yojana (BSKY):**
  - Free health coverage & cashless treatment in govt. and empanelled private hospitals.

    - Emergency patients transported under this scheme.
- **Gopabandhu Jana Arogya Yojana (GJAY):**
    - Annual coverage up to ₹5 lakh per family (extra ₹5 lakh for women after exhausting limit).
    - Used for hospitalization & emergency medical treatment.
- **Odisha State Treatment Fund (OSTF):**
    - Financial assistance up to ₹3 lakh for costly treatment.
    - Additional help possible with special approval.

Free Medicine & Diagnostics

- **Niramaya Scheme:**
    - ~570 essential medicines supplied free of cost.
    - Available at government hospitals & drug distribution centers (many open 24×7).
- **Free Diagnostics:**
    - Government facilities provide several diagnostic tests free for emergency & routine cases.

Expansion & Modernization

- Procurement of **new ambulances** (ALS, BLS, boat ambulances) to strengthen coverage.
- Plan to add **1200+ ambulances in 4 years** to reduce response time.
- Replacement of **428 old ambulances** recently approved.

- **Helipad facilities** being developed in some medical colleges for airlifting critical patients.
- Move to **regulate private ambulances** and integrate them with apps (to avoid overcharging and improve availability).

In short, Odisha provides **free ambulance services (108/Janani Express/Boat), financial aid (BSKY, GJAY, OSTF), free medicines (Niramaya), and diagnostics** — plus continuous expansion with new vehicles and facilities to ensure citizens get timely emergency medical help.

Here are sample healthcare-related questions with emotional and friendly answers based on the provided document:

Question 1: What are the symptoms of chikungunya, and how can I manage the joint pain?

Answer:

Chikungunya often starts with sudden high fever and can bring along severe joint pain, especially in wrists, ankles, fingers, and toes. It can be quite tough, especially when the joint pain lasts for weeks, making movement difficult. Remember, you are not alone—many have faced this, and resting well with plenty of fluids like coconut water helps a lot. Simple home remedies like turmeric and ginger can ease inflammation and pain. Please take it slow and reach out to your healthcare provider if the pain becomes overwhelming. Your body is

healing, and gentle care will support your strength every step of the way.

Question 2: How can I protect my family from diseases spread by mosquitoes like dengue and malaria?

Answer:

Protecting your loved ones starts at home by making small but powerful changes. Remove any stagnant water around your house, use mosquito nets especially at night, and encourage everyone to wear full-sleeved clothing. It might seem like simple steps, but they make a huge difference. Using natural mosquito repellents can keep those tiny threats at bay too. Know that caring for your family this way is a loving and powerful action—you're creating a safe environment. And if anyone feels unwell, please seek medical advice early so they can get the care they deserve.

Question 3: What can I do if my child has diarrhea, and I'm worried about dehydration?

Answer:

It can be really frightening to see your little one uncomfortable. The most important thing is to keep them hydrated—offer plenty of fluids like oral rehydration solution (ORS), coconut water, or warm lemon

water with a pinch of salt and sugar. You can also soothe their tummy with gentle foods like bananas, rice, applesauce, and toast—the BRAT diet. Remember to wash your hands thoroughly before feeding and keep their surroundings clean. If your child seems very weak, or has fewer wet diapers than usual, don't hesitate to get medical help. Your care and the love you offer is the greatest comfort for them during this time.

Question 4: I'm feeling anxious and overwhelmed due to health worries. Where can I find support?

Answer:

Your feelings are completely valid, and it's okay to seek help. Many people experience anxiety, especially when dealing with health concerns. In Odisha, you're not alone—there are organizations and helplines ready to listen and support you. Sometimes talking to a trusted friend, family member, or counselor can lift a heavy burden. Try practicing deep breathing, gentle yoga, or meditation to calm your mind. Remember, reaching out is a sign of strength. Your mental well-being is just as important as physical health, and you deserve compassion and care through this journey.

Question 5: What are some simple things I can do at home to help manage diabetes?

Answer:

Managing diabetes can feel challenging, but small daily habits can bring big positive changes. Drinking ajwain water or aloe vera juice, eating nutrient-rich foods like bitter gourd and amla, and including fenugreek seeds in your diet have helped many people. Regular, gentle exercises like walking or yoga also benefit your blood sugar and heart health. Most importantly, be gentle with yourself—this is a journey, not a race. Celebrate each step you take, and always keep in touch with your healthcare provider. You're doing great by taking charge of your health.

Question 6: I feel scared about getting vaccinated. Is it really safe?

Answer:

Your feelings are understandable—many have similar concerns. Vaccinations are one of the most effective ways to protect you and your loved ones from serious diseases. They prepare your immune system gently without causing illness, which is a kind of shield. If you have any questions or worries, speaking with a healthcare provider can help ease your mind. Remember, getting vaccinated is an act of love for yourself and your community.

Question 7: What should I do if someone in my family has tuberculosis? Is it contagious?

Answer:

Tuberculosis can feel intimidating, but you're not alone. TB spreads through the air when someone coughs or sneezes, so it's important to keep windows open for fresh air and ensure the person receives proper treatment. Encourage them with kindness—it's a long journey but very treatable with medicine. Your care and support mean a lot to their recovery. Don't hesitate to reach out to healthcare services for guidance.

Question 8: How can I protect my kids during the rainy season when diseases spread fast?

Answer:

The rainy season can be challenging, but you're doing a wonderful job by seeking ways to protect your little ones. Make sure they drink clean, boiled water and eat freshly cooked food. Keep their hands clean, nails short, and clothes dry. Using mosquito repellents and nets safeguards against bites. Your attentiveness creates a safe and loving environment that helps them stay happy and healthy.

Question 9: I'm feeling tired and weak after recovering from dengue. Is this normal?

Answer:

Absolutely, feeling tired after dengue is very common. Your body has been working hard fighting infection, and it needs time to rebuild strength. Rest as much as you can, drink nourishing fluids, and eat nutritious foods when you feel ready. Be patient with yourself—healing is a gentle process. Share your worries with family and friends; they'll support you as you regain your energy.

Question 10: What mental health support is available if I'm feeling overwhelmed by illness in my family?

Answer:

It can be tough when someone you care about is unwell, and your feelings are important. In Odisha, several helplines and counseling services are ready to listen whenever you need someone to talk to. Don't hesitate to seek help or share your emotions with trusted people around you. Remember, taking care of your own mental health is vital to support your family effectively. Small moments of self-care can help you stay strong through challenging times.

Question 11: How can I keep my elderly parents safe from infectious diseases?

Answer:

Taking care of your elderly parents during times of illness requires extra love and caution. Encourage frequent hand washing, balanced nutrition, and regular medical checkups. Their immune systems might

be weaker, so protecting them from crowded places and ensuring they get their vaccinations is very important. Your care is their greatest shield—every small step you take means the world to their health and happiness.

Question 12: What should I do if I see symptoms of cholera in my child?

Answer:

Seeing your child unwell can be very scary. If you notice watery diarrhea or vomiting, start giving oral rehydration solution immediately to prevent dehydration. Keep your child hydrated with clean water or safe fluids and consult a healthcare provider without delay. You're doing a great job by acting quickly—remember, early treatment can save lives.

Question 13: I'm worried about the stigma attached to diseases like tuberculosis. How can I handle it?

Answer:

It's heartbreaking that stigma sometimes comes with illness, but remember, TB is just a disease—nothing to be ashamed of. Educate your community gently about how TB spreads and how treatment works. Your courage to speak openly can break barriers and help others seek help too. You're building hope, and that is incredibly powerful.

Question 14: How do I stay healthy during the monsoon when many diseases spread?

Answer:

Monsoon can bring lots of challenges, but staying vigilant helps protect you and your family. Drink only boiled or purified water, avoid street food, and keep your surroundings clean to prevent mosquito and waterborne diseases. Wearing covered clothes and using mosquito nets are simple yet effective. Your mindful care nurtures health and happiness through the season.

Question 15: Can breastfeeding help protect my baby from infections?

Answer:

Absolutely! Breastfeeding provides your baby with essential nutrients and antibodies that help fight infections naturally. It’s a wonderful way to bond and give your baby the best start in life. If you ever face challenges, remember help is available from healthcare workers and support groups who understand your journey.

Question 16: What should someone do to recover well after dengue fever?

Answer:

Recovery from dengue takes time—be patient and gentle with yourself. Keep drinking lots of fluids, rest fully, and eat nutritious, easily digestible foods like fruits and soups. Avoid strenuous activities until you feel strong again. Reach out if your symptoms worsen; remember, support from family and friends makes healing smoother.

Question 17: My family lives in a remote area. How can we get reliable health information?

Answer:

Living remotely can be tough, but technology and community health workers are bridges to good healthcare information. Your AI chatbot can answer many questions anytime, and local health camps bring doctors closer to you. Keep in touch with trusted health workers, and don’t hesitate to share your concerns. You’re important, and help is reaching out to you.

Question 18: Is it normal to feel tired after recovering from malaria?

Answer:

Yes, feeling tired after malaria is very common as your body rebuilds strength. Rest is your best friend now, along with plenty of fluids and

nourishing food. Light walks or gentle stretching can help regain energy slowly. Remember, this phase is temporary—each day brings you closer to full health.

Question 19: How can I encourage my community to participate in a vaccination drive?

Answer:

Encouraging vaccination starts with sharing truthful, simple information about its benefits and safety. Share stories of health and happiness after vaccination, and listen patiently to concerns without judgment. Your kindness and knowledge can inspire trust and protect many lives—you're a vital voice for health!

Question 20: I feel lonely taking care of a sick family member. What can I do?

Answer:

Caring for someone you love can sometimes feel heavy and isolating. It's important to take moments for yourself—short walks, deep breaths, or talking to a friend. There are support groups and helplines where you can share your feelings. Remember, taking care of yourself helps you care better for others. You're doing a beautiful, brave thing, and you deserve kindness too.

Vaccination programs :

The vaccination programs in Odisha are part of a broader national initiative, the **Universal Immunization Programme (UIP)**, which is one of the world's largest public health interventions.[1] Odisha's government, through its Health & Family Welfare Department and National Health Mission, is actively working to ensure high vaccination coverage across the state.

Here is a detailed overview of the ongoing vaccination programs in Odisha:

1. Universal Immunization Programme (UIP)[2]

The UIP is the cornerstone of Odisha's vaccination efforts. It provides free-of-cost vaccines against a number of diseases to children and pregnant women.[3] The state's goal is to provide efficient and safe immunization services to all eligible individuals.

Vaccines Provided Under UIP:

- **For Pregnant Women:** Tetanus and adult Diphtheria (Td) vaccine.[4]
- **For Infants and Children:** The program covers a comprehensive schedule of vaccines, including:
    - BCG (Bacillus Calmette-Guerin) for tuberculosis.[5]
    - Hepatitis B (birth dose).[6]
    - Oral Polio Vaccine (OPV) and Inactivated Polio Vaccine (IPV).
    - Pentavalent vaccine (protects against Diphtheria, Pertussis, Tetanus, Hepatitis B, and Haemophilus Influenzae type B).

- Rotavirus vaccine (RVV) for rotavirus diarrhea.[7]
- Measles and Rubella (MR) vaccine.
- Japanese Encephalitis (JE) vaccine in endemic districts.[8]
- Pneumococcal Conjugate Vaccine (PCV) for pneumonia and meningitis.[9]
- Vitamin A supplementation.[10]

Latest Details:

- The state is focused on strengthening its immunization system to reach every child and pregnant woman.
- There is a strong collaboration with Integrated Child Development Services (ICDS) and Anganwadi workers to improve outreach, especially in remote areas.[11]
- The state is implementing "fixed-day fixed-site" strategies to ensure predictable and accessible vaccination services.[12]
- **Intensified Mission Indradhanush (IMI):** Odisha participates in national drives like IMI, which focuses on areas with low immunization coverage and aims to vaccinate children who have been left out or have dropped out of the routine program.[13] The IMI 5.0 drive has been particularly significant in targeting measles and rubella, with the aim of eliminating these diseases.[14]

2. Digital Initiatives: U-WIN Platform[15]

To improve the efficiency and tracking of the immunization program, the government of India, and consequently Odisha, has implemented a new digital platform called **U-WIN**.[16]

- **What it is:** U-WIN is a digital network that digitizes the entire vaccination process, from beneficiary registration to stock management and temperature tracking of vaccines.[17]
- **Benefits:**
    - **Beneficiary Registration:** Citizens can self-register through the U-WIN web portal or mobile app.[18]
    - **Real-time Tracking:** It allows for real-time tracking of vaccination status for children up to five years of age and pregnant women.[19]
    - **Automated Alerts:** The system sends automated SMS alerts for upcoming vaccine doses.[20]
    - **Digital Certificates:** It generates a universal QR-based eVaccination Certificate.
    - **Health Records:** It provides the option to create an Ayushman Bharat Health Account (ABHA) ID for comprehensive digital health management.[21]

3. Special Vaccination Drives

In addition to the routine immunization program, the government conducts special drives and campaigns to address specific health challenges.[22]

- **Measles and Rubella (MR) Elimination:** Odisha has intensified its MR immunization drive to achieve a target of 95% coverage in every district. Districts have been categorized as high, medium, and low priority to focus resources where they are most needed.[23] The state has been actively using the U-WIN platform to monitor this campaign in real-time.

- **COVID-19 Vaccination:** While the major COVID-19 vaccination drive has largely concluded, the government maintains a readiness to manage any new developments. Bhubaneswar became the first city in India to achieve 100% vaccination coverage for its adult population, showcasing the state's effective public health response during the pandemic.

Key Challenges and Ongoing Efforts:

Despite the successes, the state government is continually working to address challenges to ensure 100% immunization coverage. These include:

- **Reaching remote and hard-to-reach areas.**[24]
- **Tackling vaccine hesitancy**, especially in some indigenous and rural communities.
- **Ensuring a consistent supply chain** for vaccines, which is being addressed by the Electronic Vaccine Intelligence Network (eVIN) system.
- **Improving data management and monitoring** to identify and cover all unvaccinated or partially vaccinated children.

Chikungunya

- **Vaccines available / status**
  • Several chikungunya vaccines reached late clinical trials in 2021–2024 (e.g., inactivated and live-attenuated candidates from Bharat Biotech, Valneva, others). One recently marketed product (Bavarian Nordic / other vendors) received regulatory

approvals in some countries. However regulatory positions have been changing as safety reports emerged.

- Typical immune response / efficacy
  • Phase 2/3 trials show strong antibody responses and expected clinical protection, but final population-level effectiveness varies by product and is still being monitored in rollout.
- Common side-effects / safety signals
  • Usual mild local/systemic events: injection-site pain, fatigue, headache, fever. There have been a small number of serious adverse event reports prompting regulatory review in some jurisdictions (see news below) — these are being investigated.
- Recovery / impact
  • There isn't a single global "vaccine recovery rate" — trials report % reduction in symptomatic infection. Data show promising protection, but long-term monitoring is required.
- Recent news (important)
  • There have been regulatory updates: e.g., some chikungunya vaccines received approvals, while at least one (Valneva's Ixchiq) faced a U.S. license suspension after rare serious adverse event reports — regulators are investigating. This changing regulatory picture is why local health authorities emphasize post-market safety monitoring.

Dengue

- Vaccines available / status
  • TAK-003 (Qdenga, Takeda) — WHO prequalified (May 2024) and recommended for programmatic use in certain age groups in high-transmission settings (children ≈6–16 yrs; guidance

depends on serostatus and local epidemiology).
• **Dengvaxia (Sanofi)** — previously licensed in many countries; use is restricted (higher risk if given to seronegative children) and many countries follow specific serostatus guidance.

- **Typical immune response / efficacy**
  • TAK-003 showed broad protection across serotypes in trials with good efficacy against symptomatic dengue in children; WHO SAGE guidance recommends use in high-burden settings with programmatic precautions. Exact % efficacy depends on age and previous dengue exposure.

- **Common side-effects / safety**
  • Injection-site pain, fever, headache, fatigue; for Dengvaxia the main programmatic safety issue is increased risk of severe dengue among vaccinees who are seronegative at first dose — hence the need to follow WHO/SAGE age/serostatus guidance.

- **Recovery / impact**
  • Vaccination reduces symptomatic disease and hospitalizations; real-world impact depends on coverage and baseline seroprevalence. WHO prequalification allows procurement via UNICEF/Gavi.

- **Recent news**
  • WHO prequalified TAK-003 in 2024, which broadened vaccine options globally and increased procurement possibilities.

Malaria

- **Vaccines available / status**
  • **RTS,S/AS01 (Mosquirix)** — WHO recommendation (from 2021) for use in children in areas with moderate to high transmission.

• **R21/Matrix-M** — WHO recommended (Oct 2023) as a second malaria vaccine option for children; several countries have begun pilot rollouts and WHO/partners are supporting shipments to high-burden countries.

- **Typical immune response / efficacy**
  • RTS,S ~30–40% reduction in clinical malaria episodes in large trials (better at preventing severe disease). R21 has reported efficacy in the 70%+ range in seasonal administration trials — WHO judged both safe and effective enough for childhood programs in endemic areas. Real-world program impact depends on coverage and integration with nets, chemoprevention, etc.
- **Common side-effects / safety**
  • Injection site pain, fever, transient irritability; rare serious adverse events are under surveillance in rollout programs. Overall safety profile judged acceptable by WHO.
- **Deaths / impact**
  • Malaria still causes hundreds of thousands of deaths annually (≈597k deaths in 2023). Vaccines are a major new tool to reduce child mortality in Africa when combined with other measures.
- **Recent news**
  • WHO and partners have begun national rollouts and large shipments of R21 to high-burden countries (e.g., early deliveries to South Sudan); supply and implementation scale-up are ongoing.

Japanese encephalitis (JE)

- **Vaccines available / status**
  • Several licensed JE vaccines in global use: inactivated vaccines

(e.g., IXIARO) and live-attenuated SA14-14-2 (widely used in Asia). WHO and national programs use these routinely in endemic areas.

- Typical immune response / efficacy
  • High seroconversion rates and strong protection against clinical JE after the full schedule; many countries include JE vaccine in childhood immunization where JE is endemic.
- Common side-effects / safety
  • Local soreness, low-grade fever, headache; serious events are rare. WHO has safety reviews and endorses existing schedules.

Cholera

- Vaccines available / status
  • WHO-prequalified **oral cholera vaccines (OCVs)**: Dukoral, Shanchol, Euvichol (and optimized versions). They're used for outbreak control and in endemic settings; UNICEF/Gavi support procurement for campaigns.
- Typical immune response / efficacy
  • Killed OCVs (Shanchol/Euvichol) provide ~50–65% protection for 2–5 years (two-dose schedules); single-dose regimens give short-term protection useful in outbreaks. Dukoral gives good short-term protection (≈2 years).
- Common side-effects / safety
  • Mild GI symptoms (abdominal pain, diarrhea, nausea), fever, headache; overall safety profile is good and benefits in outbreaks outweigh these effects.
- Deaths / impact
  • Cholera can cause rapid dehydration and death if not treated;

OCV campaigns and rapid case management (ORS/IV fluids) dramatically cut fatalities in outbreaks.

- Recent news
  • OCV supply has been expanded in recent years with new production and prequalification steps to meet outbreak demand.

Diarrheal diseases (general / Rotavirus)

- Vaccines available / status
  • **Rotavirus vaccines** (Rotarix, RotaTeq, ROTAVAC, ROTASIIL, etc.) are used worldwide and are part of many national infant schedules. They prevent the most severe rotavirus gastroenteritis.
- Typical immune response / efficacy
  • In trials and real-world studies the vaccines reduce hospitalizations for severe rotavirus diarrhea substantially (effectiveness varies by region — higher in high-income settings, lower but still important in low-income settings).
- Common side-effects / safety
  • Mild GI upset, irritability; a small increased risk of intussusception (rare) — benefits (preventing many hospitalizations/deaths) outweigh the small risk. Surveillance is routine.
- Deaths / impact
  • Rotavirus vaccination has driven big declines in diarrhea hospitalizations and deaths among young children where implemented.

Tuberculosis (TB)

- Vaccines available / status
  • **BCG** (Bacillus Calmette-Guérin) — widely used at birth in many countries; protects well against severe childhood forms (meningitis, disseminated TB) though protection against pulmonary TB in adults is variable.
  • **M72/AS01E** — investigational vaccine that showed ~50% efficacy in a Phase IIb trial for preventing active pulmonary TB in IGRA-positive adults; Phase 3 trials and rollout studies are underway. WHO and developers consider it a leading candidate.
- Typical immune response / efficacy
  • BCG: strong protection against severe infant TB (60–80% for some forms), but variable for adult pulmonary TB. M72/AS01E: ~50% efficacy in the pivotal trial (in adults with latent infection).
- Common side-effects / safety
  • BCG: local swelling, sometimes a small ulcer at injection site; serious events are rare (mostly in immunocompromised). M72 trials report reactogenicity typical of adjuvanted subunit vaccines (injection-site reactions, transient fever) and are being closely monitored in Phase 3.
- Deaths / impact
  • TB remains a leading infectious killer — ~1.25 million deaths in 2023. Better vaccines (if proven in phase 3 and rolled out) could substantially reduce mortality.
- Recent news
  • Large Phase-3 trials (and accelerated funding / recruitment) for M72/AS01E and other new TB vaccine candidates were active in 2024–2025; MTBVAC and other candidates are in development.

Hepatitis

- **Vaccines available / status**
  • **Hepatitis A** vaccine – safe, widely used for travelers and in national programs in some countries.
  • **Hepatitis B** vaccine – safe, extremely effective; routine infant vaccination prevents chronic HBV infection (given at birth + follow-up doses).
  • **Hepatitis C – no vaccine yet** (treatment for HCV is highly effective with DAAs).
- **Typical immune response / efficacy**
  • Hep B vaccine: often >95% protective when full course completed. Hep A: strong antibody response and long-term protection after full course.
- **Common side-effects / safety**
  • Injection-site soreness, mild fever, headache; serious reactions are rare. Hep B vaccine has very good long-term safety record.
- **Recent news**
  • Ongoing public-health pushes to increase hepatitis B vaccination and screening because chronic HBV remains a major cause of liver cancer and death; hepatitis C treatment access has improved but vaccine research continues.

Leptospirosis

- **Vaccines available / status**
  • **Human leptospirosis vaccines** exist only in a few countries (e.g., China, Japan, Cuba, France) and are generally multivalent killed whole-cell vaccines targeting dominant local serovars —

they are **not** widely used globally. Many countries rely on prevention (avoid contaminated water, protective equipment) and animal vaccination.

- **Typical immune response / efficacy**
  • Licensed human vaccines provide protection against vaccine-matched serovars; cross-protection across all serovars is limited. Efficacy varies by product and region (some studies report high efficacy for specific vaccines).
- **Common side-effects / safety**
  • Mild local/systemic reactions; safety profiles differ by product; large-scale global safety data are limited because use is geographically limited.

Deaths & big picture numbers (selected high-impact figures)

- **Malaria:** ~597,000 deaths in 2023 (global estimate).
- **Tuberculosis:** ≈1.25 million deaths in 2023.
- **Dengue, cholera, hepatitis, JE, leptospirosis, chikungunya:** these cause significant regional morbidity and mortality; numbers vary year to year and by region. (WHO/Ministry of Health country reports give the best local counts.) For example, cholera outbreaks can cause high case-fatality in unprotected areas and OCV campaigns are used to prevent deaths. Dengue causes thousands of deaths annually in high-burden countries, and prequalified vaccines are a new tool.